CS 6515/4540 (Fall 2026)

Exam 2 - PRACTICE

Name: _____

GTID: _____

GT Username: _____

INSTRUCTIONS

1. When the exam starts, write your name (or GT ID or GT Username) on each page.
2. Bringing two pages of notes is permitted.
3. No books, calculators, or other electronic devices permitted.
4. Make your answers clear and concise.
5. ONLY write on the **FRONT** of each sheet. Backs will not be scanned.
6. You may refer to standard algorithms from class without giving full pseudocode.

I am aware of and in accordance with the Academic Honor Code of Georgia Tech and the Georgia Tech Code of Conduct. I will use no person's help on this test. Also, I have read all the instructions on this page.

Signature:

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1 Short Answer Questions (36 points, each part 6 points)

Unless explicitly mentioned, a short 1–2 line justification is required.

1. **LP.** Every convex set can be written as an intersection of a finite number of half-spaces (linear inequalities).

   ◯ True

   ◯ False

   False. A ball, for example, needs infinitely many half-spaces.

2. **Facility Location.**

   Consider the LP relaxation for the facility location problem from class, where I is the set of potential facility locations and J is the set of clients such that $I \cap J = \emptyset$, i.e., no client share a location with a potential facility:

   $$\min \sum_{i \in I} f_i y_i + \sum_{i \in I} \sum_{j \in J} d_{ij} x_{ij}, \quad \text{s.t. } \sum_{i \in I} x_{ij} = 1 \; \forall j \in J, \quad x_{ij} \leq y_i \; \forall i, j, \quad x_{ij}, y_i \geq 0.$$

   Suppose all opening costs are zero, i.e., $f_i = 0$ for all i. What happens to the optimal LP value?

   (A) The optimal LP value is 0.

   (B) Every client connects to its nearest facility.

   (C) The LP becomes unbounded.

   (D) The LP value equals the total connection cost of the MST.

   Answer: (B). When opening facilities is free, the optimal solution opens every facility needed to minimize connection costs. Each client connects to the nearest open facility.

3. **Farkas' Lemma.** Let $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, $b = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$. Use Farkas to show $Ax = b, x \geq 0$ is infeasible by giving a suitable λ_1, λ_2.

   Take $\lambda = (1, 1)$. Then $\lambda^\top A = (0, 0) \geq 0$ and $\lambda^\top b = -2 < 0$, which certifies infeasibility.

4. **Gradient Descent** When running gradient descent on a β-smooth, convex function, we will converge to the optimum regardless of step size or initial point chosen

   ◯ True

   ◯ False

False, we need step-size to be small enough. For instance, consider the function $f(x) = \frac{1}{2}x^2$, which is 1-smooth. Start at $x_0 = 1$ with step size 2. $x_1 = 1 - 2 = -1$, and $x_2 = -1 + 2 = 1$, so we will go back and forth between these two points infinitely.
Also, an optimum may not exist, for instance if $f(x) = x$.

5. **Convex Bodies** We can run Gradient Descent to solve LPs in polynomial time, assuming the feasible region of the LP is a **bounded** convex set and the LP objective is a linear function, which is convex.

   ◯ True

   ◯ False

False. Even if we start in the feasible region, there's no reason to think the next step keeps us in the feasible region.

6. **Approximation Algorithms**

A 2-approximation algorithm is applied to a maximization problem, and the optimal solution is known to be 5. Which of these are possible values the approximation algorithm may return? **Choose all that apply.**

A. 2.5

B. 5

C. 7

D. 10

E. None of the above.

A or B.

2 Linear Programming (25 points)

1. **Polygon (10 points).** Let P be a convex polygon given by

$$a_i x + b_i y \leq c_i \quad (i = 1, \ldots, m).$$

Given a fixed point $q = (x_0, y_0)$, find $p = (x_1, y_1) \in P$ minimizing

$$\|p - q\|_\infty = \max\{|x_1 - x_0|,\ |y_1 - y_0|\}.$$

Formulate a Linear Program that computes the minimum L_∞ distance from q to P.
(*Hint:* Introduce t with $|x_1 - x_0| \leq t$, $|y_1 - y_0| \leq t$ and linearize.)

$$\min t$$
$$t \geq x_1 - x_0$$
$$t \geq x_0 - x_1$$
$$t \geq y_1 - y_0$$
$$t \geq y_0 - y_1$$
$$a_i x_1 + b_i y_1 \leq c_i \quad (i = 1, \ldots, m).$$

2. **Matching.** Consider the (non-bipartite) maximum matching LP relaxation on $G = (V, E)$ with $\delta(v)$ denoting all the edges incident to any vertex $v \in V$:

$$\max \sum_{e \in E} x_e \quad \text{s.t.} \quad \sum_{e \in \delta(v)} x_e \leq 1 \ \forall v \in V, \qquad x_e \geq 0 \ \forall e \in E.$$

(a) LP Relaxation (5 points) Prove that for any graph $G = (V, E)$, the optimal value of this LP relaxation is at least the size of the maximum matching in G.

We let $x_e = 1$ if e is in the matching and 0 otherwise. This is feasible: clearly the degree of every vertex is ≤ 1 in a matching, thus $\sum_{e \in \delta(v)} x_e = \deg(v) \leq 1$. Hence all constraints are satisfied and we know the objective function $\sum_e x_e$ is the size of the matching. Since integral solutions are a subset of the fractional solutions, the optimal value of the LP relaxation is at least the size of the maximal matching in G

(b) Integrality gap (5 points) Give an example of a graph G where the LP relaxation value is strictly more than the size of the maximum matching on G.

(*Hint:* There is a graph already on just 3 vertices.)

Take a triangle on 3 vertices. Integral optimum picks 1 edge. LP can set $x_e = \frac{1}{2}$ on all three edges in triangle, yielding $\frac{3}{2}$.

(c) Dual of the LP (5 points). Write the dual of the above max-matching LP relaxation.

$$\min \sum_{v \in V} y_v \quad \text{s.t.} \quad y_u + y_v \geq 1 \ \forall \{u, v\} \in E, \qquad y_v \geq 0 \ \forall v \in V.$$

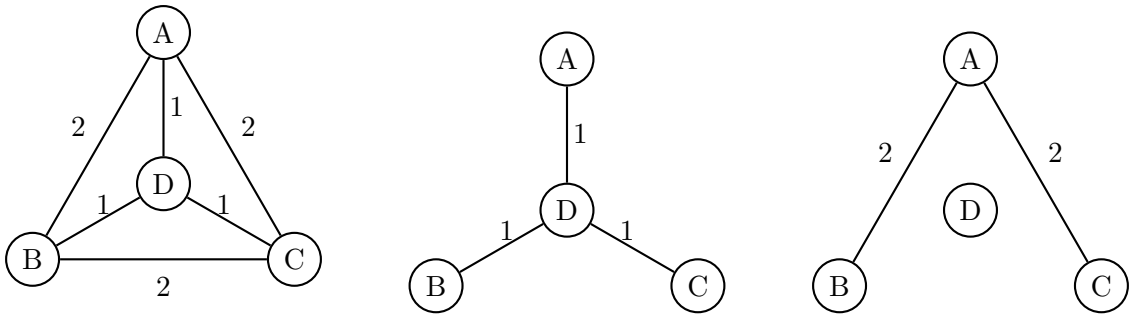
This is the fractional vertex cover LP.

3 Steiner Tree (19 Points)

Suppose we are given a fully-connected (complete) graph $G = (V, E)$ with nonnegative edge weights satisfying the triangle inequality, and a subset $T \subseteq V$ of vertices that must be connected. Find a minimum-weight subset of edges $F \subseteq E$ that connects all vertices in T (vertices in $V \setminus T$ may or may not be connected).
(*Note:* The optimal solution is a tree because otherwise we can drop a cycle edge while maintaining connectivity. Finding the minimum such tree (known as Steiner tree) is NP-complete.)

Let G_T be a fully-connected subgraph of G induced by T (i.e., the subgraph where we remove vertices $V \setminus T$). In this problem we will prove that the minimum spanning tree (MST) of G_T is a 2-approximation to the minimum

Example.



Graph with $T = \{A, B, C\}$ *Steiner Tree* MST of G_T

1. (12 points) Let F^* denote the subset of edges that form the minimum Steiner tree. Prove how we can use it to form a spanning tree of G_T (note this spanning tree cannot visit any vertex in $V \setminus T$ since G_T only has vertices T) of total weight at most twice the total weight of F^*.

   (*Hint:* The proof is similar to why MST is a 2-approx for TSP. Recall that G is complete and edge-weights satisfy triangle inequality.)

   We first construct a tour/walk of every vertex in G that visits with total weight at most twice the total weight of F^*. This can be achieved by running DFS on G using only the edges in F^*, and adding an edge to the path when you go from some u to v, as well as when you come back. For instance, in the picture above, if we break ties alphabetically, we will do $A \to D \to B \to' D \to C \to' D \to' A$, where the steps marked with an apostrophe mean you are returning back to an edge (closing that DFS subtree). Note that we use each edge of F^* exactly twice, so the total weight of this walk is double the total weight of F^*.

   Now, we can make this tour/walk into a path on vertices of T by short-circuiting. Call the walk we constructed w. Now, we can construct a path by just taking the first occurrence of each vertex of T in w. For instance, in our example, we do $A \to B \to C$, since the first occurrence of A comes before any other vertex in T, followed by B, and followed by C. By the triangle inequality, going directly from a vertex $u \to v$ is at least as good as taking walk from the first occurrence u to the first occurrence v that appears in w. So, this has total cost at most the cost of the original cost, which is at most twice the cost of F^*. It is also a spanning tree, since it is a path through every vertex in T.

2. (7 points) Using part 1 (even if you have not proved it), deduce that MST of G_T is a 2-approx for minimum Steiner tree.

We know that $2 \times w(\text{Steiner tree}) \geq w(\text{spanning tree of part (a)}) \geq w(\text{MST})$ and we know that this MST is a feasible solution as it connects all vertices in T, so the MST is a 2-approx.

4 Convex Optimization (20 points)

Definition 1 (strong convexity)**.** *A differentiable f is α-strongly convex, for $\alpha > 0$, if for all $x, y \in \mathbb{R}^d$,*

$$\boxed{f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2}$$

Consider an α-strongly convex and β-smooth function $f : \mathbb{R}^d \to \mathbb{R}$ with a global minimum at some $x^* \in \mathbb{R}^d$. We run gradient descent with no constraints (no projection) on f beginning from some point x_0, with step size $\eta = \frac{1}{\beta}$. Let $x_k \in \mathbb{R}^d$ be the iterate after k steps:

$$x_{k+1} = x_k - \frac{1}{\beta} \nabla f(x_k).$$

1. (8 points) Using the definition of β-smoothness and step size $\eta = 1/\beta$, show that

$$\frac{1}{2\beta} \|\nabla f(x_k)\|^2 \ \leq \ f(x_k) - f(x_{k+1}).$$

This is exactly the descent lemma proved in lecture (for step size $\eta = 1/\beta$): using β-smoothness, we have

$$f(x_{k+1}) \ \leq \ f(x_k) + \nabla f(x_k)^\top (x_{k+1} - x_k) + \frac{\beta}{2} \|x_{k+1} - x_k\|^2.$$

Plugging in $x_{k+1} - x_k = -\frac{1}{\beta} \nabla f(x_k)$ gives

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{\beta} \|\nabla f(x_k)\|^2 + \frac{\beta}{2} \left\| \frac{1}{\beta} \nabla f(x_k) \right\|^2 = f(x_k) - \frac{1}{2\beta} \|\nabla f(x_k)\|^2,$$

which rearranges to the desired inequality.

2. (6 points) For any α-strongly convex function f with global minimum at x^*, it can be shown that

$$\|\nabla f(x_k)\|^2 \ \geq \ 2\alpha \big(f(x_k) - f(x^*)\big).$$

Assuming this and part 1, prove that

$$f(x_{k+1}) - f(x^*) \ \leq \ \left(1 - \frac{\alpha}{\beta}\right) \big(f(x_k) - f(x^*)\big).$$

Part 1 gives

$$f(x_k) - f(x_{k+1}) \;\geq\; \frac{1}{2\beta}\|\nabla f(x_k)\|^2 \;\geq\; \frac{\alpha}{\beta}(f(x_k) - f(x^*)).$$

Rewriting,

$$f(x_{k+1}) - f(x^*) \;\leq\; (f(x_k) - f(x^*)) - \frac{\alpha}{\beta}(f(x_k) - f(x^*)) = \left(1 - \frac{\alpha}{\beta}\right)(f(x_k) - f(x^*)).$$

3. (6 points) Using the results above show that for any $k \geq 0$

$$f(x_k) - f(x^*) \leq \left(1 - \frac{\alpha}{\beta}\right)^k (f(x_0) - f(x^*)),$$

i.e., gradient descent on α-strongly convex and β-smooth functions converges exponentially fast.

From the previous part,

$$f(x_{k+1}) - f(x^*) \leq \left(1 - \frac{\alpha}{\beta}\right) (f(x_k) - f(x^*)).$$

Iterating this inequality:

$$f(x_k) - f(x^*) \leq \left(1 - \frac{\alpha}{\beta}\right)^k (f(x_0) - f(x^*)),$$

which shows exponential (linear) convergence.

END OF EXAM